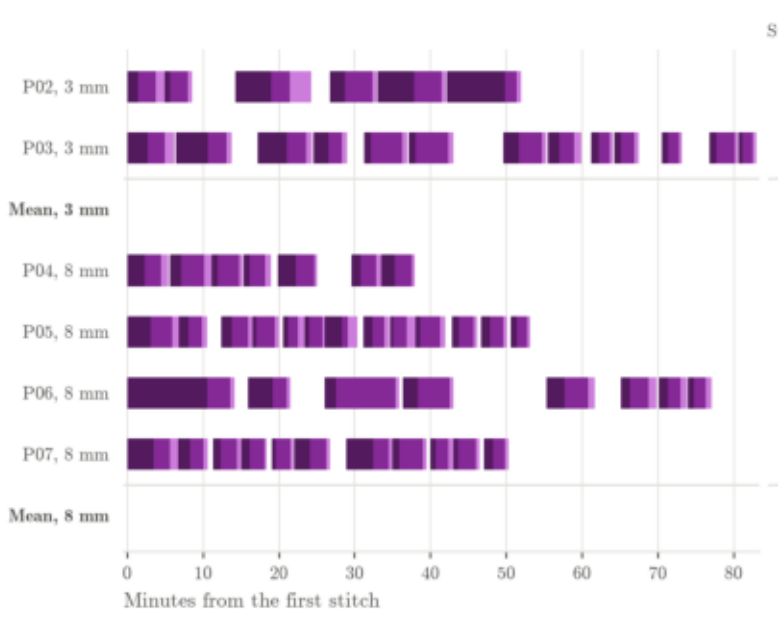
Case	Knots tied	Stitch (min)	Knot tying (min)	Assistant (s)	Force, mean (N, est.)	Force, peak (N, est.)
P02, 3 mm	6	6.2	2.7	59	1.06	2.22
P03, 3 mm	13	4.1	2.4	33	0.96	2.21
Mean, 3 mm	**10**	**5.2**	**2.5**	**46**	**1.01**	**2.22**
P04, 8 mm	7	3.9	2.4	34	1.39	3.72
P05, 8 mm	13	3.0	1.9	38	1.54	3.70
P06, 8 mm	8	6.1	3.2	38	1.74	4.15
P07, 8 mm	11	3.6	2.1	28	1.83	3.71
Mean, 8 mm	**10**	**4.1**	**2.4**	**34**	**1.63**	**3.82**

Knots tied: stitches with a completed knot. Stitch: needle first touches tissue to knot tightened. Knot tying: needle through to knot tightened. Assistant: knot tightened to the assistant's cut. Force (estimate, preliminary): insertion-axis joint effort minus a 60 s rolling median, mean and 95th percentile over each stitch, larger of the two arms; no force sensor. P01 (3 mm) is excluded. Bold rows: mean of the cases in each system.



	Stitches	Stitch (min)	Gap (min)	Span (min)
P02, 3 mm	6	6.2	2.8	52
P03, 3 mm	13	4.1	2.4	83
Mean, 3 mm	**10**	**5.2**	**2.6**	**67**
P04, 8 mm	7	3.9	1.6	38
P05, 8 mm	13	3.0	1.2	53
P06, 8 mm	8	6.1	4.0	77
P07, 8 mm	11	3.6	1.0	50
Mean, 8 mm	**10**	**4.1**	**2.0**	**55**